

Homework 12

Solution

1. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} \frac{|x|y}{|x|+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

- (a) Determine whether f is continuous at $(0, 0)$.

Solution:

Note that for every $(x, y) \neq (0, 0)$ we have

$$0 \leq |f(x, y)| = \frac{|x|}{|x|+y^2} \cdot |y| \leq |y|$$

and thus, by the squeeze rule, we have $\lim_{(x,y) \rightarrow (0,0)} |f(x, y)| = 0$, implying that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$.

- (b) Compute $f_x(0, 0)$ and $f_y(0, 0)$ or prove that they don't exist.

Solution:

Note that

$$f_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x - 0} = \lim_{x \rightarrow 0} \frac{0 - 0}{x - 0} = 0$$

and similarly

$$f_y(0, 0) = \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y - 0} = \lim_{y \rightarrow 0} \frac{0 - 0}{y - 0} = 0$$

- (c) Compute $D_{\vec{u}}(f)(0, 0)$ for the direction vector $\vec{u} = (1, 1)$.

Solution:

We note that $\|\vec{u}\| = \sqrt{2}$. Define $\vec{v} = \frac{\vec{u}}{\|\vec{u}\|}$. Then

$$D_{\vec{u}}(f)(0, 0) = D_{\vec{v}}(f)(0, 0) = \lim_{h \rightarrow 0} \frac{f\left(\frac{h}{\sqrt{2}}, \frac{h}{\sqrt{2}}\right) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h \cdot |h|}{2}}{h\left(\frac{|h|}{\sqrt{2}} + \frac{h^2}{2}\right)} = \lim_{h \rightarrow 0} \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}} + \frac{|h|}{2}} = \frac{1}{\sqrt{2}}$$

2. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

(a) Determine whether f is continuous.

Solution:

We compute

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{xy(x^2 - y^2)}{x^2 + y^2} = \lim_{r \rightarrow 0^+} \frac{r \cos \theta r \sin \theta (r^2 \cos^2 \theta - r^2 \sin^2 \theta)}{r^2} \\ &= \lim_{r \rightarrow 0^+} r^2 \cos \theta \sin \theta (\cos^2 \theta - \sin^2 \theta) \stackrel{B \times V}{=} 0 = f(0, 0) \end{aligned}$$

where we have used the fact that $\cos \theta \sin \theta (\cos^2 \theta - \sin^2 \theta)$ is bounded, since for every θ ,

$$|\cos \theta \sin \theta (\cos^2 \theta - \sin^2 \theta)| \leq |\cos^2 \theta - \sin^2 \theta| \leq |\cos^2 \theta| + |\sin^2 \theta| \leq 2$$

Thus f is continuous at $(0, 0)$. For every $(x, y) \neq (0, 0)$, f is continuous by AOC.

Thus, f is a continuous function.

(b) Compute f_x and f_y for every $(x, y) \in \mathbb{R}^2$.

Solution:

We first use the definition to compute $f_x(0, 0)$,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

For every $(x, y) \neq (0, 0)$, we can use AOD:

$$f_x(x, y) = \frac{(3x^2y - y^3)(x^2 + y^2) - xy(x^2 - y^2) \cdot 2x}{(x^2 + y^2)^2}$$

We now compute f_y .

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

For every $(x, y) \neq (0, 0)$, we can use AOD:

$$f_y(x, y) = \frac{(x^3 - 3xy^2)(x^2 + y^2) - xy(x^2 - y^2) \cdot 2y}{(x^2 + y^2)^2}$$

- (c) Compute $f_{xy}(0,0)$ and $f_{yx}(0,0)$.

Solution:

We use the definition at the origin:

$$f_{xy}(0,0) = \lim_{h \rightarrow 0} \frac{f_x(0,h) - f_x(0,0)}{h} = \lim_{h \rightarrow 0} \frac{-\frac{h^5}{h^4} - 0}{h} = \lim_{h \rightarrow 0} \frac{-h^5}{h^5} = -1$$

and

$$f_{yx}(0,0) = \lim_{h \rightarrow 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^5}{h^4} - 0}{h} = \lim_{h \rightarrow 0} \frac{h^5}{h^5} = 1$$

- (d) Prove or disprove the following statement: Both mixed derivatives f_{xy} and f_{yx} are continuous at $(0,0)$.

Solution:

False. Assume towards a contradiction that both mixed derivatives f_{xy} and f_{yx} are continuous at $(0,0)$.

Then by a theorem learned in class,

$$-1 = f_{xy}(0,0) = f_{yx}(0,0) = 1$$

Contradiction.

3. Compute the following limit, or prove that it doesn't exist in the extended sense:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2}$$

Solution:

The limit does not exist. Indeed, along the path $y = 0$ we get

$$\lim_{x \rightarrow 0} \frac{x^3 \cdot 0}{x^6 + 0^2} = \lim_{x \rightarrow 0} \frac{0}{x^6} = \lim_{x \rightarrow 0} 0 = 0$$

On the other hand, along the path $y = x^3$ we get

$$\lim_{x \rightarrow 0} \frac{x^3 \cdot x^3}{x^6 + x^6} = \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2}$$

Since the two paths give different limits, the limit does not exist.

4. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + xy + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

(a) Determine whether f is continuous at $(0, 0)$.

Solution:

Note that

$$\lim_{(x,y) \rightarrow (0,0), x=y} f(x, y) = \lim_{x \rightarrow 0} f(x, x) = \lim_{x \rightarrow 0} \frac{x^2}{3x^2} = \frac{1}{3} \neq f(0, 0)$$

and therefore f is not continuous at $(0, 0)$.

(b) Compute the value of $f_x(0, 0)$ or prove that it doesn't exist.

Solution:

We have

$$f_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x - 0} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0$$

(c) Let $a, b \in \mathbb{R}$ such that $a^2 + b^2 = 1$ and let $\vec{u} = (a, b)$. Suppose that $D_{\vec{u}}(f)(0, 0)$ exists.

Prove that $a = 0$ or $b = 0$.

Solution:

Assume towards a contradiction that $a \neq 0$ and $b \neq 0$. Set

$$r = \frac{ab}{a^2 + ab + b^2}.$$

Since $a^2 + ab + b^2 = \left(a + \frac{b}{2}\right)^2 + \frac{3b^2}{4} > 0$ and $ab \neq 0$, then r is well-defined and $r \neq 0$. Hence

$$D_{\vec{u}}(f)(0, 0) = \lim_{h \rightarrow 0} \frac{f(ah, bh) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{ab}{a^2 + ab + b^2} = \lim_{h \rightarrow 0} \frac{r}{h},$$

which does not exist. This contradicts the assumption that $D_{\vec{u}}(f)(0, 0)$ exists.

5. Prove or disprove each of the following statements:

(a) There exists a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$D(f)(x, y) = \left(x^2y + e^{xy}, \frac{x^3}{3} + xe^{xy} \right)$$

for every $(x, y) \in \mathbb{R}^2$.

Solution:

False.

Assume towards a contradiction that such a function f exists. Then for every $x, y \in \mathbb{R}$ we have

$$f_x(x, y) = x^2y + e^{xy}$$

and

$$f_y(x, y) = \frac{x^3}{3} + xe^{xy}.$$

We note that for every $x, y \in \mathbb{R}$ we have

$$f_{xy}(x, y) = \frac{\partial}{\partial y} (x^2y + e^{xy}) = x^2 + xe^{xy},$$

$$f_{yx}(x, y) = \frac{\partial}{\partial x} \left(\frac{x^3}{3} + xe^{xy} \right) = x^2 + e^{xy} + xye^{xy}.$$

We note that f_{xy} and f_{yx} are continuous by AOC and therefore $f_{xy}(x, y) = f_{yx}(x, y)$ for every $x, y \in \mathbb{R}$.

However

$$f_{xy}(0, 0) = 0, \quad f_{yx}(0, 0) = 1.$$

Contradiction.

(b) There exists a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$D(f)(x, y) = \left(x^2 y + e^y + 2x, \frac{x^3}{3} + x e^y + \cos y \right)$$

for every $(x, y) \in \mathbb{R}^2$.

Solution:

True. We choose

$$f(x, y) = \frac{x^3 y}{3} + x e^y + x^2 + \sin y, \quad \forall x, y \in \mathbb{R}$$

Note that

$$f_x(x, y) = x^2 y + e^y + 2x, \quad f_y(x, y) = \frac{x^3}{3} + x e^y + \cos y, \quad \forall x, y \in \mathbb{R}$$

and therefore

$$D(f)(x, y) = \left(x^2 y + e^y + 2x, \frac{x^3}{3} + x e^y + \cos y \right), \quad \forall (x, y) \in \mathbb{R}^2$$

(c) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function for which $f(x, y) = f(y, x)$ for every $(x, y) \in \mathbb{R}^2$.

If $f_x(1, 2)$ exists, then $f_y(2, 1)$ exists and $f_x(1, 2) = f_y(2, 1)$.

Solution:

True. We compute

$$f_x(1, 2) = \lim_{x \rightarrow 1} \frac{f(x, 2) - f(1, 2)}{x - 1} = \lim_{x \rightarrow 1} \frac{f(2, x) - f(2, 1)}{x - 1} = \lim_{y \rightarrow 1} \frac{f(2, y) - f(2, 1)}{y - 1} = f_y(2, 1)$$

(d) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function for which $D(f)(x, y)$ exists for every $(x, y) \in \mathbb{R}^2$.

Then f is continuous at every $(x, y) \in \mathbb{R}^2$.

Solution:

False. Consider the function defined by

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}, \quad \forall (x, y) \in \mathbb{R}^2$$

We have seen in class that f is not continuous at $(0, 0)$ and that $D(f)(0, 0) = (0, 0)$.

For $(x, y) \neq (0, 0)$, note also that

$$f_x(x, y) = \frac{y(y^2 - x^2)}{(x^2 + y^2)^2}, \quad f_y(x, y) = \frac{x(x^2 - y^2)}{(x^2 + y^2)^2}.$$

At $(0, 0)$, both partial derivatives exist and equal 0. Thus $D(f)(x, y)$ exists for every $(x, y) \in \mathbb{R}^2$.